

The world's first successful human kidney transplantation was performed from one twins to antoher by Joseph E. Murray and his colleagues at Peter Bent Brigham Hospital, Boston in 1954. The first ever human kidney transplant performed in India was done at the King Edward Memorial Hospital at Mumbai in May 1965, using a cadaver donor in a non-renal failure patient who had had hypernephroma. The first successful live donor kidney transplant in India was done at Christian Medical College Hospital, Vellore in January 1971 by Dr. Johnny and Dr. Mohan Rao.

Summary

Epithelial tissues are the interface between internal fluids and the external environment, creating osmotic barriers in lower organisms. Other specialized epithelial tissues that mediate osmotic and ionic regulation are gills, digestive tract and specialized excretory tissues in different animal groups. Animals remove toxic ammonia to less toxic forms by excretion. Three main strategies of nitrogen excretion are ammonoteles (ammonium), Uricoteles (uric acid) and Ureotelism (Urea). Most aquatic animals are ammonotelic, whereas terrestrial animals are uricotelic (reptiles and birds) or ureoteles (mammals). Urea is produced by the Ornithine cycle/Urea cycle in the liver.

Invertebrates have primitive kidneys such as protonephridia and metanephridia. Water balance in insects is regulated by Malphigian tubules. Ion and water regulation in vertebrates are carried out by the kidneys. The functional units of kidney is the nephron. Urine is formed by 3 processes, Glomerular filtration, tubular reabsorption and tubular secretion. Filtration occurs at the glomerulus, a ball of capillaries surrounded by the Bowman's capsule. From the Bowman's capsule the primary urine enters the proximal tubule, and proceeds to the loop of Henle, with its ascending and descending limbs. The hypertonic fluid then flows to the distal tubule and through the collecting duct into the ureters, the urinary bladder, after a short storage it is sent out of the urethra. Central to the nephron is the counter current system set up between the loop of Henle and the collecting duct along with the capillaries that serve the nephron.

Kidney function is regulated at different levels. GFR is affected by colloidal osmotic pressure and capsular hydrostatic pressure between the glomerulus and Bowman's capsule, surface area available for filtration are the factors that affect filtration pressure. The kidneys act only on the plasma, yet the extra cellular fluid consists of both plasma and interstitial fluid. The interstitial fluid is the true internal fluid environment of the body. Interstitial fluid is the only component that comes in direct contact with the cells. Thus by performing regulatory and excretory roles on the plasma, the kidneys maintains the proper interstitial fluid environment for optimum cell functioning.

Various hormones control diuresis. Vasopressin alters the permeability of the collecting duct, the renin- angiotensin system, sympathetic system and aldosterone act together to regulate $Na^{+,}$ K^{+}, water and pressure balance.

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